Anisotropic Linearization and a Frozen Minimax Exponent for Darcy Coefficient Recovery

Open-weight synthesis article

Abstract

We study the statistical inverse problem of recovering a Darcy conductivity f from Gaussian white-noise observations of the potential u_f solving $\operatorname{div}(f\nabla u) = 1$ with zero boundary data on the unit ball $D \subset \mathbb{R}^d$, $d \geq 2$. We prove the linearization at the constant conductivity, including the exact radial identity and the anisotropic frozen symbol. We show that the natural fixed-anchor quadratic remainder estimate at $f \equiv 1$ is false and prove the correct moving-anchor quadratic estimate. For a frozen periodic linearized model with nonzero frozen direction, we prove the minimax L^2 exponent

$$n^{-2\alpha/(2\alpha+d+3)}$$

(up to logarithmic factors in the certified statement; the periodic wavelet construction below gives the sharper constant-order version). We also prove that the phase-space variance of the genuine variable-coefficient linearized symbol has order R^{d+3}, and we prove a rigorous two-point lower bound at the critical point $x = 0$ giving the weaker exponent

$$n^{-(2\alpha+d)/(2\alpha+d+4)}.$$

The full nonlinear variable-coefficient minimax rate over the original Darcy class remains open.

1 Introduction and certified scope

Let

$$D = \{x \in \mathbb{R}^d : |x| < 1\}, \qquad d \geq 2.$$

For a uniformly positive conductivity f, let $u_f = \Phi(f)$ denote the solution of Darcy's problem

$$\operatorname{div}(f\nabla u) = 1 \quad \text{in } D, \qquad u|_{\partial D} = 0. \tag{1}$$

The observation model is Gaussian white noise:

$$Y_n = u_f + \frac{1}{\sqrt{n}}\,\mathbb{W},$$

where $\mathbb{W}$ is Gaussian white noise on $L^2(D)$. The parameter class is

$$\mathcal{F}(\alpha, r) = \left\{ h \in C^\alpha(\overline{D}) : \|h - \mathbf{1}\|_{C^\alpha(\overline{D})} \le r \right\},$$

with $\alpha > 1$ and fixed $r \in (0, 1)$. The original minimax question is the asymptotic size, up to logarithmic factors, of

$$\inf_{\widehat{f}_n} \sup_{f_0 \in \mathcal{F}(\alpha,r)} \mathbb{E}_{f_0} \left[\left\| \widehat{f}_n - f_0 \right\|^2_{L^2(D)} \right].$$

This paper proves a scoped set of results that identify the linearized anisotropy and the resulting frozen minimax exponent, while leaving the full nonlinear variable-coefficient problem open. The main certified results are:

(i) the linearization at $f \equiv 1$, the exact radial identity, and injectivity of the linearized operator on regular functions;

(ii) the failure of the fixed-anchor quadratic estimate at $f \equiv 1$, and the validity of the moving-anchor quadratic estimate;

(iii) the minimax rate of a frozen periodic linearized model with nonzero frozen direction, giving the exponent $2\alpha/(2\alpha + d + 3)$;

(iv) a phase-space variance calculation for the variable-coefficient linearized symbol, also of order R^{d+3};

(v) a critical-point two-point lower bound for the original nonlinear problem, giving the weaker exponent $(2\alpha + d)/(2\alpha + d + 4)$.

The frozen exponent is the candidate exponent suggested by the anisotropic linearization. The critical-point lower bound is strictly weaker and shows that the origin, although more degenerate locally, is not the minimax bottleneck.

Throughout, $A \asymp B$ means $cA \le B \le CA$, with constants depending only on the structural parameters indicated in the statement. Constants denoted by C, c may change from line to line.

2 Preliminaries

We use the weak formulation of (1). For $f \in L^\infty(D)$ with $f \geq \lambda > 0$, $u_f \in H_0^1(D)$ is the unique solution of

$$\int_D f \nabla u_f \cdot \nabla \varphi \, dx = -\int_D \varphi \, dx \qquad \forall \varphi \in H_0^1(D). \tag{2}$$

Let C_P denote the Poincaré constant of D.

Lemma 2.1 (Forward well-posedness and uniform energy bound)**.** *Let $f \in L^\infty(D)$ and $f \geq \lambda > 0$. Then (2) has a unique solution $u_f \in H_0^1(D)$, and*

$$\|\nabla u_f\|_{L^2(D)} \leq \frac{C_P}{\lambda}, \qquad \|u_f\|_{L^2(D)} \leq \frac{C_P^2}{\lambda}.$$

Proof. The bilinear form

$$B(u, \varphi) = \int_D f \nabla u \cdot \nabla \varphi \, dx$$

is continuous and coercive on $H_0^1(D)$, since

$$B(\varphi, \varphi) \geq \lambda \|\nabla \varphi\|_{L^2}^2 .$$

The right-hand side $\varphi \mapsto -\int_D \varphi$ is bounded on $H_0^1(D)$ by Poincaré's inequality. Existence and uniqueness follow from the Lax–Milgram theorem; see [1].

Testing (2) with $\varphi = u_f$ gives

$$\int_D f |\nabla u_f|^2 \, dx = -\int_D u_f \, dx.$$

Hence

$$\lambda \|\nabla u_f\|_2^2 \leq \|u_f\|_2 \leq C_P \|\nabla u_f\|_2 ,$$

so $\|\nabla u_f\|_2 \leq C_P/\lambda$. A second application of Poincaré gives the L^2 bound. □

For $f \geq \lambda > 0$ and $h \in L^\infty(D)$, define the linearized operator $A_f h$ weakly by

$$\int_D f \nabla (A_f h) \cdot \nabla \varphi \, dx = -\int_D h \nabla u_f \cdot \nabla \varphi \, dx \qquad \forall \varphi \in H_0^1(D). \tag{3}$$

Again by Lax–Milgram, $A_f h \in H_0^1(D)$ is well-defined whenever $h\nabla u_f \in L^2(D)$; in particular this holds for $h \in L^\infty(D)$. On the regular positive cone this operator is the Fréchet derivative of the forward map Φ.

We shall use the standard Cameron–Martin formula for Gaussian white noise. If P_s denotes the law of $s + n^{-1/2}\mathbb{W}$, then for $s, t \in L^2$,

$$\mathrm{KL}(P_s \| P_t) = \frac{n}{2} \|s - t\|_{L^2}^2 .$$

For completeness we record this as a lemma.

Lemma 2.2 (KL divergence in Gaussian white noise)**.** *Let $\{e_k\}_{k\geq 1}$ be an orthonormal basis of $L^2(D)$. Under the white-noise model, the coordinates*

$$Y_k = \langle Y_n, e_k \rangle$$

are independent Gaussian random variables with variance $1/n$ and means $\langle s, e_k \rangle$. Therefore, for $s, t \in L^2(D)$,

$$\mathrm{KL}(P_s \| P_t) = \sum_{k\geq 1} \frac{n}{2} \big(\langle s - t, e_k \rangle\big)^2 = \frac{n}{2} \|s - t\|_{L^2(D)}^2 .$$

Proof. For one coordinate, $Y_k \sim N(\langle s, e_k \rangle, 1/n)$ under P_s and $Y_k \sim N(\langle t, e_k \rangle, 1/n)$ under P_t. The KL divergence between two Gaussian laws with equal variance σ^2 is $(\mu - \nu)^2/(2\sigma^2)$. Here $\sigma^2 = 1/n$, hence the k-th contribution is

$$\frac{n}{2} \big(\langle s, e_k \rangle - \langle t, e_k \rangle\big)^2.$$

Summing over k and applying Parseval gives the claim. □

For Hölder–Zygmund spaces we use the standard norm $C^\alpha(\overline{D})$, including the supremum norm. For non-integer α, C^α coincides with the Besov space $B^\alpha_{\infty,\infty}$; see [2]. In the frozen periodic model we use the Besov formulation directly.

3 Linearization at the constant conductivity

Proposition 3.1 (Forward solution and derivative at $f = 1$)**.** *Let*

$$u_1(x) = \frac{|x|^2 - 1}{2d}.$$

Then $\Phi(1) = u_1$, $\Delta u_1 = 1$, $\nabla u_1 = x/d$, *and* $u_1|_{\partial D} = 0$. *For smooth* h, *the Fréchet derivative* $A = A_1 = \Phi'(1)$ *is given by*

$$Ah = w, \qquad \Delta w = -\left(\frac{x}{d} \cdot \nabla h + h\right), \qquad w|_{\partial D} = 0.$$

Equivalently,

$$-\Delta w = \frac{x}{d} \cdot \nabla h + h.$$

If $h \equiv g$ *is constant, then*

$$Ag = -gu_1.$$

Proof. A direct calculation gives

$$\Delta u_1 = \frac{1}{2d}\Delta(|x|^2) = 1,$$

and $u_1 = 0$ on $|x| = 1$. Hence u_1 solves (1) for $f = 1$.

Let $f_\varepsilon = 1 + \varepsilon h$ and write

$$u_{f_\varepsilon} = u_1 + \varepsilon w + o(\varepsilon)$$

in the weak sense. The weak equation is

$$\int_D (1 + \varepsilon h)\nabla u_{f_\varepsilon} \cdot \nabla \varphi \, dx = -\int_D \varphi \, dx.$$

Taking the first-order term in ε gives

$$\int_D \nabla w \cdot \nabla \varphi \, dx + \int_D h \nabla u_1 \cdot \nabla \varphi \, dx = 0.$$

Since $\nabla u_1 = x/d$, this is the weak form of

$$-\Delta w = \operatorname{div}\left(h\frac{x}{d}\right) = \frac{x}{d} \cdot \nabla h + h.$$

Equivalently,

$$\Delta w = -\left(\frac{x}{d} \cdot \nabla h + h\right).$$

If $h \equiv g$, then the right-hand side is $-g$. Since $\Delta u_1 = 1$ and $u_1|_{\partial D} = 0$, the unique solution is $w = -gu_1$. □

Proposition 3.2 (Exact radial identity)**.** *Let $h(x) = \phi(|x|)$ be radial and sufficiently smooth. Then $w = Ah$ is radial. Writing $w(x) = W(\rho)$, $\rho = |x|$, one has*

$$W'(\rho) = -\frac{\rho}{d}\phi(\rho), \qquad 0 < \rho < 1.$$

Equivalently,

$$W(\rho) = \int_\rho^1 \frac{s}{d}\phi(s)\, ds.$$

Proof. By rotational invariance and uniqueness, w is radial. The radial Laplacian is

$$\Delta w = W''(\rho) + \frac{d-1}{\rho}W'(\rho).$$

For $h(x) = \phi(\rho)$,

$$\frac{x}{d} \cdot \nabla h + h = \frac{\rho}{d}\phi'(\rho) + \phi(\rho).$$

Thus W satisfies

$$W'' + \frac{d-1}{\rho}W' = -\frac{\rho}{d}\phi' - \phi.$$

If $W'(\rho) = -(\rho/d)\phi(\rho)$, then

$$W'' = -\frac{1}{d}\phi - \frac{\rho}{d}\phi',$$

and

$$\frac{d-1}{\rho}W' = -\frac{d-1}{d}\phi.$$

Adding gives

$$-\frac{1}{d}\phi - \frac{\rho}{d}\phi' - \frac{d-1}{d}\phi = -\phi - \frac{\rho}{d}\phi',$$

which is exactly the radial equation. Regularity at $\rho = 0$ removes the singular integration constant, and $W(1) = 0$ yields the integral representation. □

Proposition 3.3 (Injectivity of A on regular functions)**.** *If $h \in C^\alpha(\overline{D})$, $\alpha > 1$, and $Ah = 0$, then $h = 0$.*

Proof. Since the Dirichlet Laplacian is injective, $Ah = 0$ implies

$$\frac{x}{d} \cdot \nabla h + h = 0$$

in the distributional sense. Because $\alpha > 1$, $h \in C^1$, so the identity holds classically. In polar coordinates $x = \rho\omega$,

$$\frac{\rho}{d}\partial_\rho h(\rho\omega) + h(\rho\omega) = 0,$$

or equivalently

$$\partial_\rho\big(\rho^d h(\rho\omega)\big) = 0.$$

Thus

$$h(\rho\omega) = \rho^{-d} g(\omega).$$

Since h is bounded at $\rho = 0$, necessarily $g(\omega) = 0$ for every ω. Hence $h = 0$. □

Remark 3.4. Formally freezing the linearized operator at a nonzero point x_0 with $v = x_0/d \neq 0$ gives the constant-coefficient Fourier multiplier

$$a_v(\xi) = \frac{1 + i\xi \cdot v}{|\xi|^2}.$$

The radial identity shows degree-one smoothing in radial directions, while the frozen symbol exhibits an equatorial degeneracy where $\xi \cdot v$ is small. Finite-difference computations corroborate the tangential smoothing but are not used in the proofs.

4 Anchor correction and moving-anchor stability

The natural fixed-anchor estimate with $A = A_1$ would assert that

$$\|\Phi(f) - \Phi(f') - A_1(f - f')\|_{L^2(D)} \leq C \|f - f'\|_{C^\alpha}^2$$

uniformly on $\mathcal{F}(\alpha, r)$. This is false.

Proposition 4.1 (Failure of the fixed-anchor quadratic estimate)**.** *Fix $r \in (0, 1)$. There is no finite constant C such that*

$$\|\Phi(f) - \Phi(f') - A_1(f - f')\|_{L^2(D)} \leq C \|f - f'\|_{C^\alpha}^2$$

holds for all $f, f' \in \mathcal{F}(\alpha, r)$.

Proof. It suffices to use constant conductivities. Let

$$b = 1 + \frac{r}{2}, \qquad 0 < g < \frac{r}{2},$$

and set

$$f = b, \qquad f' = b - g.$$

Both belong to $\mathcal{F}(\alpha, r)$, because their deviations from 1 are at most $r/2$. For a constant conductivity $c > 0$, uniqueness and linearity give

$$\Phi(c) = \frac{u_1}{c}.$$

By Proposition 3.1, for the constant increment g,

$$A_1 g = -g u_1.$$

Therefore

$$\Phi(f) - \Phi(f') - A_1(f - f') = u_1 \left(\frac{1}{b} - \frac{1}{b-g} + g \right).$$

But

$$\frac{1}{b} - \frac{1}{b-g} + g = g \left(1 - \frac{1}{b(b-g)} \right).$$

As $g \downarrow 0$,

$$1 - \frac{1}{b(b-g)} \longrightarrow 1 - \frac{1}{b^2} > 0,$$

because $b > 1$. Since $\|f - f'\|_{C^\alpha} = g$ for constant increments,

$$\frac{\|\Phi(f) - \Phi(f') - A_1(f - f')\|_{L^2}}{\|f - f'\|_{C^\alpha}^2} \geq \frac{c}{g} \to \infty.$$

Thus no finite constant C can exist. □

The correct estimate anchors the linearization at the base point f'.

Theorem 4.2 (Uniform moving-anchor quadratic remainder)**.** *Let $\alpha > 1$ and $r \in (0, 1)$. There exists $C = C(d, \alpha, r) < \infty$ such that for all $f, f' \in \mathcal{F}(\alpha, r)$,*

$$\|\Phi(f) - \Phi(f') - A_{f'}(f - f')\|_{L^2(D)} \leq C \, \|f - f'\|_{C^\alpha(\overline{D})}^2 .$$

One may take $C \leq C_d (1 - r)^{-3}$; in particular the constant blows up only as $r \uparrow 1$.

Proof. Write

$$g = f - f', \qquad u = \Phi(f), \qquad u' = \Phi(f'), \qquad v = A_{f'}g, \qquad R = u - u' - v.$$

Since $f, f' \in \mathcal{F}(\alpha, r)$,

$$f, f' \geq 1 - r =: \lambda > 0, \qquad \|g\|_{L^\infty} \leq \|g\|_{C^\alpha}.$$

Let $q = u - u'$. From the weak equations for u and u',

$$\int_D f' \nabla q \cdot \nabla \varphi \, dx = -\int_D g \nabla u \cdot \nabla \varphi \, dx \qquad \forall \varphi \in H_0^1(D).$$

Testing with $\varphi = q$ gives

$$\lambda \|\nabla q\|_2^2 \leq \|g\|_\infty \|\nabla u\|_2 \|\nabla q\|_2 .$$

By Lemma 2.1, $\|\nabla u\|_2 \leq C_P/\lambda$, hence

$$\|\nabla q\|_2 \leq \frac{C_P}{\lambda^2} \|g\|_\infty .$$

By definition of $v = A_{f'}g$,

$$\int_D f' \nabla v \cdot \nabla \varphi \, dx = -\int_D g \nabla u' \cdot \nabla \varphi \, dx.$$

Subtracting this from the equation for q yields

$$\int_D f' \nabla R \cdot \nabla \varphi \, dx = -\int_D g \nabla q \cdot \nabla \varphi \, dx.$$

Testing with $\varphi = R$ gives

$$\lambda \|\nabla R\|_2^2 \leq \|g\|_\infty \|\nabla q\|_2 \|\nabla R\|_2 .$$

Therefore

$$\|\nabla R\|_2 \leq \frac{1}{\lambda} \|g\|_\infty \|\nabla q\|_2 \leq \frac{C_P}{\lambda^3} \|g\|_\infty^2 .$$

Poincaré's inequality gives

$$\|R\|_2 \leq C_P \|\nabla R\|_2 \leq \frac{C_P^2}{(1-r)^3} \|g\|_\infty^2 .$$

Since $\|g\|_\infty \leq \|g\|_{C^\alpha}$, the theorem follows. □

5 Frozen periodic linearized inverse problem

We now analyze a frozen periodic linearized model. The original PDE has variable coefficients and Dirichlet boundary conditions; the frozen model is a local periodic surrogate. We choose coordinates so that the frozen nonzero direction is e_d. Let

$$\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d .$$

Fix $\beta > 0$ and define the Fourier multiplier

$$a_\beta(\xi) = \frac{1 + i\beta\xi_d}{|\xi|^2}, \qquad \xi \in \mathbb{Z}^d \setminus \{0\}.$$

Let A_β be the corresponding Fourier multiplier operator on mean-zero functions. The frozen Gaussian sequence model is

$$Y_\xi = a_\beta(\xi)\theta_\xi + n^{-1/2} Z_\xi, \qquad \xi \neq 0, \tag{4}$$

where θ_ξ are the Fourier coefficients of a real mean-zero function θ, and Z_ξ denote the Fourier coordinates of Gaussian white noise. We suppress the usual conjugacy constraints for real signals; they affect constants only. The zero mode is set to zero: $\theta_0 = 0$.

The parameter space is the Besov ball

$$\Theta_\alpha(L) = \left\{ \theta \in B^\alpha_{\infty,\infty}(\mathbb{T}^d) : \|\theta\|_{B^\alpha_{\infty,\infty}} \le L,\ \theta_0 = 0 \right\},$$

where $\alpha > 0$. We use the Littlewood–Paley characterization

$$\|\theta\|_{B^\alpha_{\infty,\infty}} \asymp \sup_{j \ge 0} 2^{j\alpha} \|\Delta_j \theta\|_{L^\infty} ;$$

see [2].

5.1 Counting and variance

Lemma 5.1 (Weyl counting for the frozen symbol)**.** *Fix $\beta > 0$. For $0 < \sigma \le \sigma_0(d, \beta)$, define*

$$N_\beta(\sigma) = \#\{\xi \in \mathbb{Z}^d \setminus \{0\} : |a_\beta(\xi)| > \sigma\}.$$

Then

$$N_\beta(\sigma) \asymp_{d,\beta} \sigma^{-d}.$$

Proof. For $|\xi| \geq 1$,

$$|a_\beta(\xi)| = \frac{\sqrt{1+\beta^2\xi_d^2}}{|\xi|^2} \leq C_\beta \frac{1+|\xi|}{|\xi|^2} \leq \frac{C_\beta}{|\xi|}.$$

Thus $|a_\beta(\xi)| > \sigma$ implies $|\xi| \leq C_\beta/\sigma$, giving

$$N_\beta(\sigma) \leq C\sigma^{-d}.$$

For the lower bound, restrict to the cone

$$|\xi_d| \geq \frac{1}{2}|\xi|.$$

On this cone,

$$|a_\beta(\xi)| \geq \frac{\beta|\xi_d|}{|\xi|^2} \geq \frac{\beta}{2|\xi|}.$$

If

$$1 \leq |\xi| \leq c\beta/\sigma,$$

then $|a_\beta(\xi)| \geq \sigma$. The number of lattice points in this cone and ball is comparable to σ^{-d}, with constants depending on d and β. Hence

$$N_\beta(\sigma) \geq c\sigma^{-d}.$$

□

The counting law suggests degree-one smoothing, but it is not the rate-determining quantity. The minimax rate is controlled by the variance sum.

Lemma 5.2 (Frozen variance sum)**.** *Fix $\beta > 0$. For $R \geq 1$,*

$$S_\beta(R) = \sum_{\substack{\xi\in\mathbb{Z}^d \\ 1\leq|\xi|\leq R}} |a_\beta(\xi)|^{-2} = \sum_{\substack{\xi\in\mathbb{Z}^d \\ 1\leq|\xi|\leq R}} \frac{|\xi|^4}{1+\beta^2\xi_d^2} \asymp_{d,\beta} R^{d+3}.$$

Proof. For the lower bound, take $\xi_d = 0$ and $\xi' = (\xi_1, \ldots, \xi_{d-1})$ in the annulus

$$R/2 \leq |\xi'| \leq R.$$

There are $\asymp R^{d-1}$ such lattice points, and for each,

$$\frac{|\xi|^4}{1+\beta^2\xi_d^2} = |\xi'|^4 \asymp R^4.$$

Thus

$$S_\beta(R) \geq cR^{d+3}.$$

For the upper bound, decompose into dyadic shells

$$2^k \leq |\xi| < 2^{k+1}, \qquad 0 \leq k \leq \lceil \log_2 R \rceil.$$

On such a shell, $|\xi|^4 \leq C2^{4k}$. Write $\xi = (\eta, m)$ with $\eta \in \mathbb{Z}^{d-1}$ and $m = \xi_d$. For fixed η with $|\eta| \leq C2^k$,

$$\sum_{m \in \mathbb{Z}} \frac{1}{1 + \beta^2 m^2} \leq C_\beta.$$

The number of such η is $\asymp 2^{k(d-1)}$. Hence the k-th shell contributes at most

$$C_\beta 2^{4k} 2^{k(d-1)} = C_\beta 2^{k(d+3)}.$$

Summing over k up to $\log_2 R$ gives

$$S_\beta(R) \leq CR^{d+3}.$$

□

Lemma 5.3 (Besov projection tail)**.** *Let $\theta \in \Theta_\alpha(L)$, and let Π_R be the Fourier projection onto $\{|\xi| \leq R\}$. Then*

$$\|\theta - \Pi_R \theta\|^2_{L^2(\mathbb{T}^d)} \leq C_{d,\alpha} L^2 R^{-2\alpha}.$$

Proof. Let Δ_j be dyadic Littlewood–Paley projections. Choose J with $2^J \leq R < 2^{J+1}$. The tail $\theta - \Pi_R \theta$ is controlled by dyadic blocks with $j \gtrsim J$. Since

$$\|\Delta_j \theta\|_{L^\infty} \leq CL2^{-j\alpha},$$

and $\mathbb{T}^d$ has finite measure,

$$\|\Delta_j \theta\|_{L^2} \leq CL2^{-j\alpha}.$$

Therefore

$$\|\theta - \Pi_R \theta\|_{L^2} \leq \sum_{j \geq J-1} \|\Delta_j \theta\|_{L^2} \leq CL \sum_{j \geq J-1} 2^{-j\alpha} \leq CL2^{-J\alpha} \leq CLR^{-\alpha}.$$

Squaring gives the claim. □

5.2 Frozen minimax theorem

Lemma 5.4 (Wavelet packing in the equatorial slab)**.** *Let $d' = d - 1$. For every sufficiently large dyadic scale j, there exist real orthonormal functions*

$$\psi_{j,1}, \dots, \psi_{j,N_j} \in L^2(\mathbb{T}^{d'})$$

with $N_j \asymp 2^{jd'}$, whose Fourier supports lie in a dyadic annulus

$$c2^j \le |\eta| \le C2^j,$$

and such that for any coefficients b_k with $|b_k| \le b$,

$$\left\| \sum_{k=1}^{N_j} b_k \psi_{j,k} \right\|_{L^\infty(\mathbb{T}^{d'})} \le Cb\, 2^{jd'/2}.$$

Extending these functions constantly in the x_d-variable gives functions

$$\Psi_{j,k}(x', x_d) = \psi_{j,k}(x')$$

on $\mathbb{T}^d$, orthonormal in $L^2(\mathbb{T}^d)$, with Fourier support in

$$\{(\eta, 0) : c2^j \le |\eta| \le C2^j\}.$$

Proof. This is the standard periodized Meyer wavelet construction on the torus; see Meyer [3, Chapter 3] and Daubechies [4, Chapter 5]. The periodized Meyer system is an orthonormal basis with Fourier support in dyadic annuli. At scale j, the number of wavelets is comparable to $2^{jd'}$, each has L^∞-size $O(2^{jd'/2})$, and the uniform overlap estimate

$$\sum_k |\psi_{j,k}(x')| \le C2^{jd'/2}$$

holds. The stated L^∞ bound follows immediately for bounded coefficients. Extension constantly in x_d preserves orthonormality and places all Fourier mass at $\xi_d = 0$. □

Lemma 5.5 (Assouad hypercube lemma)**.** *Let $\{\psi_k\}_{k=1}^N$ be an orthonormal system in a Hilbert space, and let*

$$\theta_\omega = \delta \sum_{k=1}^{N} \omega_k \psi_k, \qquad \omega \in \{\pm 1\}^N.$$

Let P_ω be probability laws of an observation under θ_ω. Suppose that whenever ω, ω' differ in exactly one coordinate,

$$\mathrm{KL}(P_\omega \| P_{\omega'}) \le \kappa.$$

If $\kappa \le 1/8$, then for every estimator $\widehat{\theta}$,

$$\sup_\omega \mathbb{E}_\omega \left\| \widehat{\theta} - \theta_\omega \right\|^2 \ge cN\delta^2$$

with an absolute constant $c > 0$.

Proof. Let Ω be uniform on $\{\pm 1\}^N$, and let $X \sim P_\Omega$. For any estimator $\widehat{\theta}$, define bit estimators

$$\widehat{\Omega}_k = \mathrm{sign}\langle \widehat{\theta}, \psi_k \rangle.$$

Since $\langle \theta_\Omega, \psi_k \rangle = \delta \Omega_k$, if $\widehat{\Omega}_k \ne \Omega_k$, then

$$|\langle \widehat{\theta} - \theta_\Omega, \psi_k \rangle| \ge \delta.$$

By orthonormality,

$$\left\| \widehat{\theta} - \theta_\Omega \right\|^2 \ge \delta^2 \sum_{k=1}^N \mathbf{1}_{\{\widehat{\Omega}_k \ne \Omega_k\}}.$$

Averaging over Ω gives

$$\mathbb{E} \left\| \widehat{\theta} - \theta_\Omega \right\|^2 \ge \delta^2 \sum_{k=1}^N \mathbb{P}(\widehat{\Omega}_k \ne \Omega_k).$$

For fixed k, the average bit error is bounded below by the Bayes error for testing the two mixture laws $P_{k,+}$ and $P_{k,-}$ obtained by conditioning on $\Omega_k = \pm 1$. Thus

$$\mathbb{P}(\widehat{\Omega}_k \ne \Omega_k) \ge \frac{1}{2} \left(1 - \mathrm{TV}(P_{k,+}, P_{k,-})\right).$$

By convexity of KL divergence and the adjacent KL assumption,

$$\mathrm{KL}(P_{k,+} \| P_{k,-}) \le \kappa.$$

Pinsker's inequality [5] gives

$$\mathrm{TV}(P_{k,+}, P_{k,-}) \le \sqrt{\kappa/2} \le \frac{1}{2}$$

when $\kappa \le 1/8$. Hence each bit error is at least $1/4$, and

$$\mathbb{E}\left\|\widehat{\theta} - \theta_\Omega\right\|^2 \ge \frac{1}{4} N\delta^2.$$

The supremum risk is at least the average risk. □

The certified frozen statement is an up-to-logarithmic-factor exponent. The periodic wavelet construction actually gives the sharper constant-order bounds stated below.

Theorem 5.6 (Frozen minimax rate)**.** *Fix $\beta > 0$, $\alpha > 0$, and $L > 0$. For the frozen periodic model* (4) *over $\Theta_\alpha(L)$, let*

$$\mathcal{R}_n^{\mathrm{fr}} = \inf_{\widehat{\theta}} \sup_{\theta \in \Theta_\alpha(L)} \mathbb{E}\left\|\widehat{\theta} - \theta\right\|_{L^2(\mathbb{T}^d)}^2.$$

Then, with

$$p = 2\alpha + d + 3,$$

there exist constants $c, C > 0$, depending only on d, α, β, L, such that for all sufficiently large n,

$$c\, n^{-2\alpha/p} \le \mathcal{R}_n^{\mathrm{fr}} \le C\, n^{-2\alpha/p}.$$

In particular, the frozen minimax exponent is $2\alpha/(2\alpha + d + 3)$; in the certified wording this is stated up to logarithmic factors, and the periodic wavelet construction removes the logarithm.

Proof. **Upper bound.** Fix $R \ge 1$ and define the sieve estimator

$$\widehat{\theta}_\xi^R = \begin{cases} Y_\xi / a_\beta(\xi), & 1 \le |\xi| \le R, \\ 0, & \text{otherwise.} \end{cases}$$

Then

$$\mathbb{E}\left\|\widehat{\theta}^R - \theta\right\|_2^2 \le C\left(n^{-1} \sum_{1 \le |\xi| \le R} |a_\beta(\xi)|^{-2} + \|\theta - \Pi_R \theta\|_2^2\right).$$

By Lemma 5.2 and Lemma 5.3,

$$\sup_{\theta \in \Theta_\alpha(L)} \mathbb{E}\left\|\widehat{\theta}^R - \theta\right\|_2^2 \le C\left(\frac{R^{d+3}}{n} + L^2 R^{-2\alpha}\right).$$

Choose

$$R \asymp (nL^2)^{1/(2\alpha+d+3)}.$$

The two terms balance and yield

$$\mathcal{R}_n^{\mathrm{fr}} \le C_{d,\alpha,\beta,L} n^{-2\alpha/(2\alpha+d+3)}.$$

Lower bound. Let $d' = d-1$ and $p = 2\alpha+d+3 = 2\alpha+d'+4$. Choose a dyadic scale $R = 2^j$ comparable to $(nL^2)^{1/p}$, and let $\{\Psi_k\}_{k=1}^N$ be the functions from Lemma 5.4 at that scale, so $N \asymp R^{d'}$. Define

$$\delta = c_0 L R^{-\alpha-d'/2},$$

with $c_0 > 0$ small. For $\omega \in \{\pm 1\}^N$, set

$$\theta_\omega = \delta \sum_{k=1}^N \omega_k \Psi_k.$$

The Fourier support of every Ψ_k lies in an annulus $|\xi| \asymp R$ with $\xi_d = 0$, so all θ_ω lie in a single Littlewood–Paley block. By Lemma 5.4,

$$\|\theta_\omega\|_{L^\infty} \le C\delta R^{d'/2} = C c_0 L R^{-\alpha}.$$

Since the block is at frequency R,

$$\|\theta_\omega\|_{B^\alpha_{\infty,\infty}} \le C R^\alpha \|\theta_\omega\|_{L^\infty} \le C c_0 L.$$

Choosing c_0 small enough gives $\theta_\omega \in \Theta_\alpha(L)$.

For adjacent ω, ω' differing in one coordinate,

$$\theta_\omega - \theta_{\omega'} = 2\delta\Psi_k$$

for some k. Since Ψ_k has Fourier support with $\xi_d = 0$ and $|\xi| \asymp R$,

$$|a_\beta(\xi)|^2 = \frac{1}{|\xi|^4} \asymp R^{-4}$$

on that support. Hence

$$\|A_\beta(\theta_\omega - \theta_{\omega'})\|_{L^2}^2 \le C\delta^2 R^{-4}.$$

By Lemma 2.2, the KL divergence between the corresponding Gaussian laws is

$$\mathrm{KL}(P_\omega \| P_{\omega'}) = \frac{n}{2} \|A_\beta(\theta_\omega - \theta_{\omega'})\|_2^2 \le Cn\delta^2 R^{-4}.$$

Using $R^p \asymp nL^2$ and $\delta^2 = c_0^2 L^2 R^{-2\alpha-d'}$, we obtain

$$n\delta^2 R^{-4} \asymp c_0^2 nL^2 R^{-p} \asymp c_0^2.$$

Choosing c_0 small makes this at most $1/8$. Lemma 5.5 gives

$$\mathcal{R}_n^{\mathrm{fr}} \ge cN\delta^2 \asymp R^{d'} L^2 R^{-2\alpha-d'} = L^2 R^{-2\alpha}.$$

Since $R \asymp (nL^2)^{1/p}$,

$$L^2 R^{-2\alpha} \asymp n^{-2\alpha/p}.$$

This proves the lower bound. □

6 Variable-coefficient phase-space variance

For the true linearization at $f = 1$, freezing at x gives the symbol

$$a(x,\xi) = \frac{1 + i(x/d)\cdot\xi}{|\xi|^2}.$$

The corresponding phase-space variance is

$$S_D(R) = \int_D \int_{|\xi|\le R} |a(x,\xi)|^{-2}\, d\xi\, dx = \int_D \int_{|\xi|\le R} \frac{|\xi|^4}{1 + |(x/d)\cdot\xi|^2}\, d\xi\, dx.$$

Lemma 6.1 (Angular integral)**.** *Let $e \in S^{d-1}$ and $t \ge 0$. Define*

$$J(t) = \int_{S^{d-1}} \frac{d\sigma(\omega)}{1 + t^2(\omega\cdot e)^2}.$$

Then

$$J(t) \asymp_d \min\{1, t^{-1}\},$$

with the convention $t^{-1} = \infty$ at $t = 0$.

Proof. Write $s = \omega \cdot e$. Then

$$J(t) = c_d \int_{-1}^{1} \frac{(1-s^2)^{(d-3)/2}}{1+t^2 s^2}\, ds$$

for $d \geq 3$, with the usual integrable endpoint weight for $d = 2$.

If $0 \leq t \leq 1$, then $1 \leq 1 + t^2 s^2 \leq 2$, so $J(t) \asymp 1$.

Let $t \geq 1$. For the lower bound, restrict to $|s| \leq 1/(2t)$. Near $s = 0$, the weight is bounded below and the denominator is at most 2. Hence

$$J(t) \geq c \int_{-1/(2t)}^{1/(2t)} ds = \frac{c}{t}.$$

For the upper bound, split into $|s| \leq 1/t$ and $|s| > 1/t$. On $|s| \leq 1/t$, the weight is integrable near 0, giving a contribution $O(1/t)$. On $|s| > 1/t$, the denominator is at least $t^2 s^2$, so the contribution is bounded by

$$t^{-2} \int_{|s|>1/t} \frac{(1-s^2)^{(d-3)/2}}{s^2}\, ds.$$

For $d \geq 3$, the weight is bounded, and

$$\int_{1/t}^{1} s^{-2}\, ds \leq Ct.$$

For $d = 2$, the weight is $(1-s^2)^{-1/2}$; near $s = 0$ the same bound gives $O(t)$, while near the endpoints the integral is $O(1)$. Hence in all cases the second contribution is $O(1/t)$. Therefore $J(t) \leq C/t$. □

Theorem 6.2 (Variable-coefficient variance order)**.** *For $d \geq 2$ and $R \geq 1$,*

$$S_D(R) \asymp_d R^{d+3}.$$

Proof. For fixed x, put $\rho = |x|/d$. By rotational symmetry of the ξ-integral,

$$\int_{|\xi|\leq R} \frac{|\xi|^4}{1+|(x/d)\cdot\xi|^2}\, d\xi = \int_0^R r^{d+3} J(\rho r)\, dr.$$

By Lemma 6.1,

$$J(\rho r) \asymp \begin{cases} 1, & \rho r \leq 1, \\ (\rho r)^{-1}, & \rho r \geq 1. \end{cases}$$

If $R\rho \le 1$, the integral is $\asymp R^{d+4}$. If $R\rho > 1$, the dominant contribution is

$$\asymp \rho^{-1} \int_{1/\rho}^{R} r^{d+2}\, dr \asymp \rho^{-1} R^{d+3}.$$

Split D into

$$D_0 = \{x : |x| \le d/R\}, \qquad D_1 = D \setminus D_0.$$

On D_0, the inner integral is $O(R^{d+4})$, while $|D_0| \asymp R^{-d}$; the contribution is $O(R^4)$, which is lower order because $d \ge 2$.

On D_1, the contribution is comparable to

$$R^{d+3} \int_{D_1} \frac{d}{|x|}\, dx.$$

Since $d \ge 2$,

$$\int_{d/R<|x|<1} \frac{dx}{|x|} = |S^{d-1}| \int_{d/R}^{1} r^{d-2}\, dr \asymp 1.$$

Thus the outer contribution is $\asymp R^{d+3}$. The lower bound also follows directly by restricting to $1/2 \le |x| \le 1$, where ρ is bounded below. Hence

$$S_D(R) \asymp R^{d+3}.$$

□

This variance order matches the frozen-model order. It supports the candidate exponent $2\alpha/(2\alpha + d + 3)$ for the original linearized operator, but by itself it is neither an estimator upper bound nor a lower-bound construction for the full variable-coefficient nonlinear problem.

7 Critical-point two-point lower bound

At the origin, $\nabla u_1(0) = 0$, and the linearized operator degenerates further. We prove a rigorous two-point Le Cam lower bound based on origin-localized radial bumps.

Fix a nonzero radial $\psi \in C_c^\infty(B(0,1))$ with $\psi \ge 0$ and $\|\psi\|_{L^2} = 1$. For $a > 0$ and $0 < \varepsilon < 1/2$, define

$$h_{\varepsilon,a}(x) = a\psi(x/\varepsilon).$$

Lemma 7.1 (Degree-two gain at the critical point)**.** *Let $A = A_1$. For radial ψ as above,*

$$\|Ah_{\varepsilon,a}\|_{L^2(D)} = a\varepsilon^{2+d/2}\big(C_\psi + o(1)\big)$$

as $\varepsilon \downarrow 0$, where $C_\psi \in (0,\infty)$ depends only on ψ. Moreover,

$$\|\nabla Ah_{\varepsilon,a}\|_{L^2(D)} \le C'_\psi a\varepsilon^{1+d/2}.$$

Proof. Set $y = x/\varepsilon$. Then

$$h_{\varepsilon,a}(x) + \frac{x}{d}\cdot\nabla h_{\varepsilon,a}(x) = a\left(\psi(y) + \frac{y\cdot\nabla\psi(y)}{d}\right) =: aS(y).$$

But

$$S(y) = \frac{1}{d}\operatorname{div}_y(y\psi(y)).$$

Let $\widehat{w}$ solve on $\mathbb{R}^d$

$$-\Delta_y\widehat{w} = S,$$

with decay at infinity. Since ψ is radial, S is radial, compactly supported in $B(0,1)$, and has mean zero:

$$\int_{\mathbb{R}^d} S(y)\,dy = \frac{1}{d}\int_{\mathbb{R}^d}\operatorname{div}(y\psi)\,dy = 0.$$

For $|y| > 1$, the radial solution satisfies

$$\widehat{w}'(r) = -r^{1-d}\int_0^r s^{d-1}S(s)\,ds.$$

For $r \ge 1$, the integral is the total mass of S, hence zero. Thus $\widehat{w}' = 0$ for $r \ge 1$, and decay forces $\widehat{w} = 0$ there. Therefore $\widehat{w}$ is compactly supported in $B(0,1)$.

Define

$$w(x) = a\varepsilon^2\widehat{w}(x/\varepsilon).$$

Then

$$\Delta_x w = a\Delta_y\widehat{w}(y) = -aS(y),$$

so w solves the Dirichlet problem for $Ah_{\varepsilon,a}$, because its support lies in $B(0,\varepsilon) \subset D$. Hence

$$\|Ah_{\varepsilon,a}\|_2^2 = a^2\varepsilon^4\int_D |\widehat{w}(x/\varepsilon)|^2\,dx = a^2\varepsilon^{4+d}\,\|\widehat{w}\|_2^2\,.$$

Set $C_\psi = \|\widehat{w}\|_2$. If $\widehat{w} = 0$, then $S = 0$, hence $\operatorname{div}(y\psi) = 0$. For smooth compactly supported radial ψ, this forces $\psi = 0$, contradicting the choice of ψ. Thus $C_\psi > 0$.

Finally,

$$\nabla w(x) = a\varepsilon \nabla \widehat{w}(x/\varepsilon),$$

so

$$\|\nabla A h_{\varepsilon,a}\|_2 = a\varepsilon^{1+d/2} \|\nabla \widehat{w}\|_2 ,$$

which gives the gradient bound. □

Lemma 7.2 (Nonlinear remainder near the critical point)**.** *Let $u_h = \Phi(1+h)$ with $h = h_{\varepsilon,a}$ as above, and let $w_1 = Ah$. Assume a is small enough that $1 + h \geq 1/2$. Then the remainder $w_2 = u_h - u_1 - w_1$ satisfies*

$$\|w_2\|_{L^2(D)} \leq C a^2 \varepsilon^{1+d/2}.$$

Consequently, if $a = c_0 r \varepsilon^\alpha$ with $\alpha > 1$, then

$$\frac{\|w_2\|_{L^2}}{\|Ah\|_{L^2}} \leq C c_0 r \, \varepsilon^{\alpha-1} \to 0.$$

Proof. Let $w = u_h - u_1$. Since u_h solves $\operatorname{div}((1 + h)\nabla u_h) = 1$ and u_1 solves $\Delta u_1 = 1$,

$$\operatorname{div}((1 + h)\nabla w) + \operatorname{div}(h\nabla u_1) = 0.$$

Weakly,

$$\int_D (1 + h)\nabla w \cdot \nabla \varphi = -\int_D h\nabla u_1 \cdot \nabla \varphi.$$

The linear part $w_1 = Ah$ satisfies

$$\int_D \nabla w_1 \cdot \nabla \varphi = -\int_D h\nabla u_1 \cdot \nabla \varphi.$$

Subtracting, $w_2 = w - w_1$ satisfies

$$\int_D (1 + h)\nabla w_2 \cdot \nabla \varphi = -\int_D h\nabla w_1 \cdot \nabla \varphi.$$

Testing with $\varphi = w_2$ and using $1 + h \geq 1/2$ gives

$$\|\nabla w_2\|_2 \leq C \|h\nabla w_1\|_2 \leq C \|h\|_\infty \|\nabla w_1\|_2 .$$

By Lemma 7.1,

$$\|\nabla w_1\|_2 \le Ca\varepsilon^{1+d/2},$$

and $\|h\|_\infty \le a$. Therefore

$$\|\nabla w_2\|_2 \le Ca^2\varepsilon^{1+d/2}.$$

Poincaré's inequality gives the L^2 bound.

By Lemma 7.1,

$$\|Ah\|_2 \asymp a\varepsilon^{2+d/2}.$$

Hence

$$\frac{\|w_2\|_2}{\|Ah\|_2} \le Ca\varepsilon^{-1} = Cc_0 r\varepsilon^{\alpha-1},$$

which tends to zero for $\alpha > 1$. □

Lemma 7.3 (Two-point Le Cam bound)**.** *Let P_0, P_1 be probability measures with $\mathrm{KL}(P_0\|P_1) \le \kappa \le 1/8$. Let f_0, f_1 be two parameters with separation $\Delta = \|f_1 - f_0\|^2$. Then*

$$\inf_{\widehat{f}} \max_{i=0,1} \mathbb{E}_i \left\|\widehat{f} - f_i\right\|^2 \ge \frac{\Delta}{16}.$$

Proof. For any estimator $\widehat{f}$, define a test by choosing the closer parameter:

$$\varphi = \begin{cases} 0, & \left\|\widehat{f} - f_0\right\| \le \left\|\widehat{f} - f_1\right\|, \\ 1, & \text{otherwise.} \end{cases}$$

If f_0 is true and $\varphi = 1$, then by the triangle inequality,

$$\sqrt{\Delta} \le \left\|\widehat{f} - f_0\right\| + \left\|\widehat{f} - f_1\right\| \le 2\left\|\widehat{f} - f_0\right\|,$$

so $\left\|\widehat{f} - f_0\right\|^2 \ge \Delta/4$. Similarly under f_1, if $\varphi = 0$, then $\left\|\widehat{f} - f_1\right\|^2 \ge \Delta/4$. Therefore

$$\max_i \mathbb{E}_i \left\|\widehat{f} - f_i\right\|^2 \ge \frac{\Delta}{8}\big(1 - \mathrm{TV}(P_0, P_1)\big).$$

Pinsker's inequality [5] gives

$$\mathrm{TV}(P_0, P_1) \le \sqrt{\kappa/2} \le \frac{1}{2},$$

and the claim follows. □

Theorem 7.4 (Critical-point lower bound for the original problem)**.** *Let $d \geq 2$, $\alpha > 1$, and fix $r \in (0,1)$. There exists $c > 0$ such that for all large n,*

$$\inf_{\widehat{f}_n} \sup_{f \in \mathcal{F}(\alpha,r)} \mathbb{E}_f \left\| \widehat{f}_n - f \right\|_{L^2(D)}^2 \geq c\, n^{-\frac{2\alpha+d}{2\alpha+d+4}}.$$

Proof. Let

$$q = 2\alpha + d + 4.$$

Choose $c_0 > 0$ small enough that

$$c_0 \|\psi\|_{C^\alpha} \leq \frac{1}{4}, \qquad c_0 \|\psi\|_\infty \leq \frac{1}{4}.$$

Define

$$\varepsilon_n = (c_1 n r^2)^{-1/q}$$

with $c_1 > 0$ large, and set

$$a_n = c_0 r \varepsilon_n^\alpha, \qquad f_1 = 1 + a_n \psi(\cdot/\varepsilon_n), \qquad f_0 = 1.$$

By the standard scaling of Hölder–Zygmund norms,

$$\|f_1 - 1\|_{C^\alpha} \leq C c_0 r \leq r$$

for c_0 small, so $f_1 \in \mathcal{F}(\alpha, r)$. Also $f_0 \in \mathcal{F}(\alpha, r)$, and $f_1 \geq 1$ because $\psi \geq 0$.

By Lemma 7.1 and Lemma 7.2, for large n,

$$\|u_{f_1} - u_{f_0}\|_{L^2} \asymp a_n \varepsilon_n^{2+d/2} = c_0 r \varepsilon_n^{\alpha+2+d/2}.$$

Thus

$$\|u_{f_1} - u_{f_0}\|_2^2 \asymp r^2 \varepsilon_n^{2\alpha+d+4}.$$

By Lemma 2.2, the KL divergence between the white-noise laws is

$$\mathrm{KL}(P_0 \| P_1) = \frac{n}{2} \|u_{f_1} - u_{f_0}\|_2^2 \asymp n r^2 \varepsilon_n^{2\alpha+d+4}.$$

By the choice of ε_n, this is bounded by a universal constant; choosing c_1 large makes it at most $1/8$.

The parameter separation is

$$\|f_1 - f_0\|_2^2 = a_n^2 \varepsilon_n^d \|\psi\|_2^2 = c_0^2 r^2 \varepsilon_n^{2\alpha+d}.$$

Substituting $\varepsilon_n = (c_1 n r^2)^{-1/q}$ gives

$$\|f_1 - f_0\|_2^2 \asymp r^2 (nr^2)^{-\frac{2\alpha+d}{2\alpha+d+4}}.$$

For fixed r, this is comparable to

$$n^{-\frac{2\alpha+d}{2\alpha+d+4}}.$$

Lemma 7.3 yields the lower bound. □

Proposition 7.5 (Comparison of exponents)**.** *For $d \geq 2$ and $\alpha > 0$, define*

$$e_{\text{crit}} = \frac{2\alpha + d}{2\alpha + d + 4}, \qquad e_{\tan} = \frac{2\alpha}{2\alpha + d + 3}, \qquad e_{\text{cand}} = \frac{2\alpha}{2\alpha + d + 2}.$$

Then

$$e_{\tan} < e_{\text{cand}} < e_{\text{crit}}.$$

Thus the critical-point lower bound is strictly weaker than the frozen/tangential exponent, and the frozen exponent is strictly worse than the naive degree-one candidate.

Proof. First,

$$e_{\text{cand}} > e_{\tan}$$

is immediate because $2\alpha + d + 2 < 2\alpha + d + 3$.

Next, with $A = 2\alpha + d$,

$$e_{\text{crit}} - e_{\tan} = \frac{A}{A+4} - \frac{2\alpha}{A+3} = \frac{A(A+3) - 2\alpha(A+4)}{(A+4)(A+3)}.$$

The numerator is

$$(2\alpha + d)(2\alpha + d + 3) - 2\alpha(2\alpha + d + 4) = 2\alpha(d-1) + d(d+3) > 0$$

for $d \geq 2$.

Finally,

$$e_{\text{crit}} - e_{\text{cand}} = \frac{A}{A+4} - \frac{2\alpha}{A+2} = \frac{A(A+2) - 2\alpha(A+4)}{(A+4)(A+2)}.$$

The numerator is

$$(2\alpha + d)(2\alpha + d + 2) - 2\alpha(2\alpha + d + 4) = 2\alpha(d-2) + d(d+2) > 0$$

for $d \geq 2$. This proves the ordering. □

8 Open points and omitted material

The original minimax rate over $\mathcal{F}(\alpha, r)$ remains open. The certified results above identify the candidate exponent

$$\frac{2\alpha}{2\alpha + d + 3}$$

from the frozen linearized model and give supporting phase-space variance for the variable-coefficient linearization, but they do not prove the full variable-coefficient nonlinear minimax theorem. The missing ingredients include:

(1) a variable-coefficient upper bound achieving the candidate exponent, using a sieve or spectral estimator together with the moving-anchor stability of Theorem 4.2;

(2) a matching lower bound for the original problem using tangential needlets with vanishing moments and a C^α-admissible taper away from the origin;

(3) rigorous transfer from the frozen periodic multiplier model to the Dirichlet ball operator, including boundary effects and spatial localization;

(4) extension beyond the critical-point two-point lower bound of Theorem 7.4.

The frozen exponent is certified up to logarithmic factors; the periodic wavelet construction gives a sharper constant-order statement only for the frozen surrogate. A Bayesian posterior-contraction direction, numerical smoothing checks, and schema/connectivity probes are omitted as unrelated to the certified core.

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